

Shangeth Rajaa

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Spoken Dialogue Systems · Speech Representations · Multimodal LLMs · Turn-Taking · Speech Understanding and Synthesis

EDUCATION

- **Birla Institute of Technology and Science (BITS Pilani), Goa Campus** Goa, India
B.E. in Electrical & Electronics Engineering and M.Sc. in Mathematics (Dual Degree) 2021

EXPERIENCE

- **Anyreach AI** Remote
Senior Machine Learning Research Scientist Feb 2025 – Present
 - End-to-end spoken dialogue systems: training, evaluation, and deployment of ASR/LLM/TTS pipelines.
 - Research and development of improving turn-taking in ASR-LLM-TTS pipelines, semantic LLMs, and acoustic models for turn detection.
 - Multimodal LLMs for speech understanding and automatic speech translation.
 - Dual-channel turn-taking modeling; DualTurn accepted at Interspeech 2026.
 - Full-duplex spoken dialogue models for TOD system.
- **ScoreTravel AI** Remote
Senior Machine Learning Researcher Jul 2024 – Feb 2025
 - Ranker algorithm design for LLM reasoning-based recommendation engine.
 - Multimodal LLM for short-form video (Reels, TikTok) understanding.
 - System Design and Development of personalized Agentic Travel booking and recommendations.
- **skit.ai (vernacular.ai)** Bangalore
Machine Learning Researcher Jun 2021 – Jul 2024
 - Speech synthesis, prosody transfer, voice cloning, prosody-aware SLU, language ID, speaker recognition and verification.
 - End-to-end spoken dialogue systems: training, evaluation, and deployment from traditional intent-based systems to LLMs.
 - Conversational agents with LLMs; LLM alignment using RLHF and DPO for goal optimization.
 - Multi-modal LLM for speech understanding and conversational agent action.
- **Speech and Language Lab, NTU Singapore** Singapore
Research Assistant Aug 2020 – Jun 2021
 - Unsupervised and semi-supervised speech and speaker representation learning.
 - Speech understanding tasks: speaker profiling (height, age, gender), accent recognition, and accented speech recognition.
- **IBM Research Labs** Remote
Research Intern May 2020 – Aug 2020
 - Developed novel quality metrics and data transformations for structured data.
 - Optimization of data quality transformations using deep reinforcement learning.
- **INRIA France** Remote
Research Intern/Collaborator Jan 2018 – Jan 2020
 - Auto deep learning project under Prof. Isabelle Guyon.
 - Multi-modal auto deep learning models and AutoDL challenge series, NeurIPS 2019.

PUBLICATIONS

- **Interspeech 2026:** DualTurn: Learning turn-taking from dual-channel generative speech pretraining. arXiv:2603.08216
- **Interspeech 2023:** Improving end-to-end SLU performance with prosodic attention and distillation. DOI:10.21437/Interspeech.2023-1760
- **ICASSP 2023:** Improving spoken language identification with Map-Mix. DOI:10.1109/ICASSP49357.2023.10095765
- **arXiv 2022:** Skit-S2I: An Indian accented speech to intent dataset. arXiv:2212.13015
- **arXiv 2021:** Learning speaker representation with semi-supervised learning for speaker profiling. arXiv:2110.13653
- **PMLR / NeurIPS 2020:** Towards automated deep learning: analysis of the AutoDL challenge series 2019. Co-authored with Zhengying Liu, Isabelle Guyon et al. *Proceedings of Machine Learning Research*.
- **ECMLPKDD Workshop 2019:** Overview and unifying conceptualization of automated machine learning. Co-authored with Zhengying Liu, Isabelle Guyon et al. *Workshop on Automating Data Science (ADS)*.
- **Springer ICACDS 2019:** Convolutional feature extraction and neural arithmetic logic units for stock prediction. Co-authored with J.K. Sahoo. *International Conference on Advances in Computing and Data Sciences*, pp. 349–359.

OTHER EXPERIENCE & COMMUNITY

RESEARCH

- **Feature extraction and arithmetic logic units for stock price prediction:** under Prof. J.K. Sahoo
- **Audio representation learning and feature disentanglement:** under Prof. Ashwin Srinivasan

SOFTWARE AND AI DEVELOPMENT

- **AI developer intern:** OpexAI — Vision ML models for self-driving cars — 2017
- **Software developer intern:** KGLLP Fintech — Finance and trading applications — 2018
- **Computer vision developer:** Science and Technology Center — Vision-based security system — 2018

COMMUNITY

- **Google AI Explore ML:** Instructor for deep learning course under Google AI Explore ML program.
- **Google Code-In:** Mentor under the TensorFlow organization.
- **OpenCV.org:** Deep learning content developer for “Deep Learning with PyTorch” course.
- **NeurIPS 2019 AutoDL competition:** Organized AutoDL competitions (AutoCV, AutoNLP, AutoSpeech) in collaboration with INRIA, Paris.

OPEN-SOURCE

- **wavencoder:** Open-source Python package for audio encoder models and transforms for deep learning tasks.
- **SpeechLLM:** Multi-modal LLM for speech understanding tasks: speech activity detection, ASR, gender, age, accent, and emotion recognition. HuggingFace
- **Wren:** Family of multi-modal LLMs (<3B params) for speech understanding, synthesis, multi-modal representation learning, and conversational agents. HuggingFace
- **Semantic turn-taking LLM:** Small-sized LLM to predict agent action (start speaking, continue listening, start listening, continue speaking) based on conversation context. HuggingFace

SKILLS

- **Languages:** Python, C/C++, JavaScript, MATLAB
- **ML frameworks:** PyTorch, TensorFlow, JAX
- **Tools & platforms:** Git, AWS, GCP, Docker, L^AT_EX, other Agentic Tools
- **Research areas:** Multi-Modal Conversation Modelling, LLM, LLM Alignment, Speech Synthesis/Understanding, Speech Representation Learning